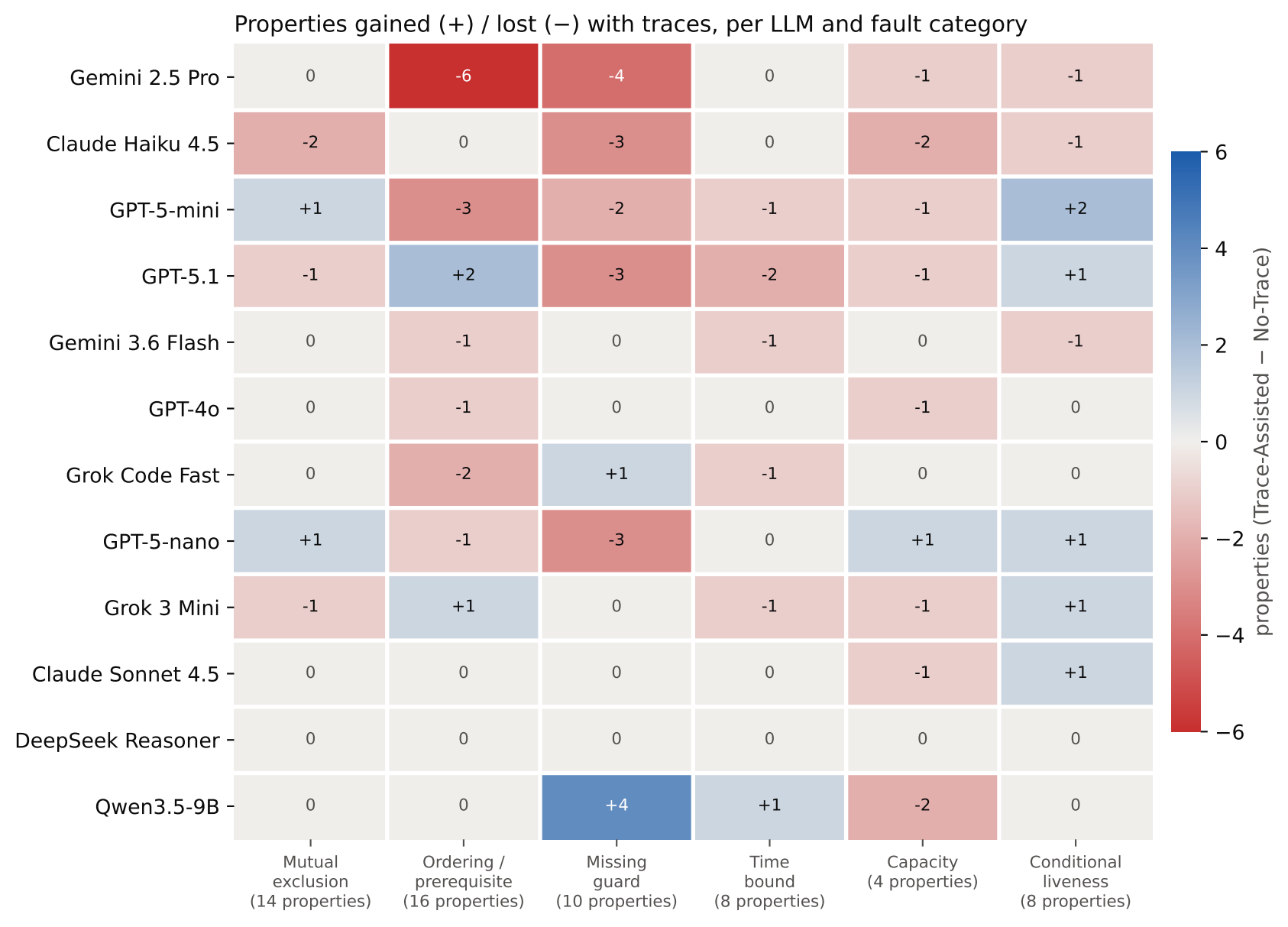
Properties gained (+) / lost (−) with traces, per LLM and fault category
Gemini 2.5 Pro
Claude Haiku 4.5
GPT-5-mini
GPT-5.1
Gemini 3.6 Flash
GPT-4o
Grok Code Fast
GPT-5-nano
Grok 3 Mini
Claude Sonnet 4.5
DeepSeek Reasoner
Qwen3.5-9B
0 -6 -4 0 -1 -1
-2 0 -3 0 -2 -1
+1 -3 -2 -1 -1 +2
-1 +2 -3 -2 -1 +1
0 -1 0 -1 0 -1
0 -1 0 0 -1 0
0 -2 +1 -1 0 0
+1 -1 -3 0 +1 +1
-1 +1 0 -1 -1 +1
0 0 0 0 -1 +1
0 0 0 0 0 0
0 0 +4 +1 -2 0
Mutual exclusion (14 properties)
Ordering / prerequisite (16 properties)
Missing guard (10 properties)
Time bound (8 properties)
Capacity (4 properties)
Conditional liveness (8 properties)
6
4
2
0
−2
−4
−6
properties (Trace-Assisted − No-Trace)